PALEOMAGNETISM OF THE MISSISSIPPIAN HP PIPE AND THE WESTERN MARGIN OF THE NORTH AMERICAN CRATON

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Abstract. The HP pipe is a small oval (80m x 40m) kimberlitic diatreme near Golden, B.C., in the fold-and-thrust belt of the Rocky Mountains. The pipe intrudes gently-dipping (~10°S) Late Cambrian to Middle Ordovician limestones with nearly vertical contacts. It is comprised of carbonate breccia fragments and pyroxenite nodules in a light-green serpentine matrix and gives a Rb-Sr age of 348 ± 7 Ma. A later phase of dark-green dikes cut the breccia and adjacent limestones. AF and thermal step demagnetization of 168 specimens from 42 oriented blocks was done using an automated cryogenic magnetometer in a shielded room for remanence measurement. The diatreme breccia and dikes retain a stable A remanence component after thermal cleaning above ~400°C and AF cleaning above ~20 mT. Their combined mean direction of 129.8°, -44.3° (α_{95} = 6.5°) after tilt correction gives a concordant Mississippian pole position of 138.5°E, 44.5°N (D_p = 5.1°, D_m = 8.2°). The limestones provide a positive contact test by giving a stable B component direction of 114.4°, -22.7° (α_{95} = 12.4°) with a concordant Late Cambrian to Middle Ordovician pole of 141.0°E, 24.3°N (δ_p = 7.0°, δ_m = 13.3°). These concordant poles indicate that the fold-and-thrust belt is part of the North American craton. Thus the suture with allochthonous terranes to the west must be located at or west of the Rocky Mountain Trench. Both the diatreme and limestones retain a low-coercivity and unblocking-temperature C component, isolated by vector subtraction without tilt correction, that records a minor overprint from the Laramide Orogeny combined with Pleistocene viscous remanence.

Introduction

Very few paleomagnetic data have been published for the fold-and-thrust belt of the Rocky Mountains in Canada (Fig. 1). Somewhere within this zone is the suture separating the North American craton from the accreted allochthonous terranes of British Columbia. The quandary is "where" and "when." For example, Stone (1977) suggested that the Rocky Mountain Trench inside the fold-and-thrust belt formed the boundary in mid-Cretaceous time. Later Monger et al. (1982) placed the suture on the western side of the belt in mid-Jurassic time as Terrane I impacted against the craton. Recently Irving et al. (1986) found anomalously low remanence inclinations in the Crowsnest volcanics which, if explained by northward translation, place the suture on the eastern side of the belt in mid-Cretaceous time. This study was undertaken for two reasons. The first was to see if the Paleozoic kimberlitic diatremes

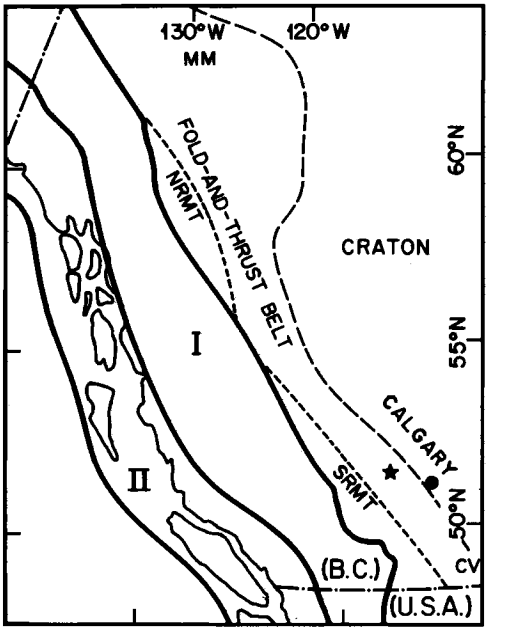


Fig. 1. Location of the Fold-and-Thrust Belt in western Canada. I and II are Terranes I and II (Monger, et al., 1982). NRMT and SRMT are the northern and southern sections of the Rocky Mountain Trench. The star locates the HP Pipe, and CV and MM locate the Crowsnest volcanics and the Mackenzie Mountains.

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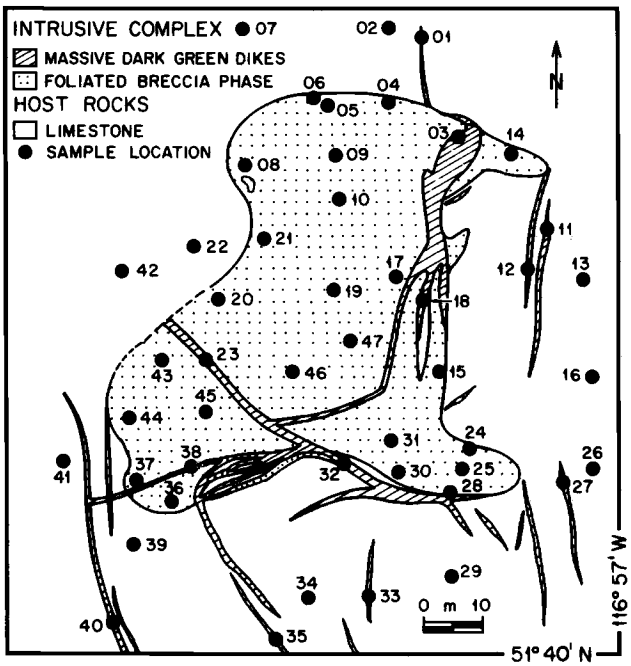


Fig. 2. Geology of the HP Pipe after Pell (1986) with site locations.

in the fold-and-thrust belt are suitable for paleomagnetic analysis, and the second was to see if they give a concordant cratonic pole position or discordant allochthonous pole position.

Geology

The HP Pipe is located about 50 km north of Golden, B.C., in the fold-and-thrust belt and east of the southern Rocky Mountain Trench (Fig. 1). The pipe is the most southerly of a small swarm of Carboniferous kimberlitic diatremes (Pell, 1986). The pipe is well exposed in a recently deglaciated valley at the terminus of the Campbell Glacier.

The HP Pipe as mapped and described by Pell (1986) is a small oval diatreme (80m x 40m) that intrudes subhorizontal (dip 0°-20°S) grey limestones of Late Cambrian to Middle Ordovician age (Price 1967) with near-vertical contacts (Fig. 2). The pipe has a dominant breccia and a subordinate dike phase. The breccia has a light-green, serpentine-rich, coarse-grained carbonate matrix with a vertical foliation that parallels the contacts. The matrix contains up to 40% inclusions of white marble, green diopside and black pyroxenite. The dike phase is dark-green, fine-grained and massive with occasional phenocrysts of pyroxene and phlogopite. These dikes are generally less than 2 m wide and are nearly vertical. They cut the breccia phase with both sharp and gradational contacts, and they extend up to 25 m into the host rock limestones with sharp contacts. Ijewliw (1986) reports that both phases contain about 1% magnetite with minor hematite. Pell (1987 and pers. comm.) notes that R. Armstrong obtained a lower Mississippian Rb-Sr age of 348 ± 7 Ma on a biotite-phlogopite separate, and that J. Harakal and K. Scott obtained Early Devonian K-Ar ages of 391 ± 12 and 396 ± 10 Ma on biotite and phlogopite respectively from the breccia phase. It is very unlikely that the Rb-Sr system would be reset to a younger age without also resetting the K-Ar system, and the initial ratio of 0.704 for the Rb-Sr determination shows no evidence of resetting. Therefore, the older K-Ar ages likely reflect excess initial argon, possibly because of two-stage cooling as the phenocrysts suggest or because of a deep-seated mantle magma source as the kimberlitic rocks suggest. No post-intrusive sedimentary rock units were deposited in the area; however, the region was structurally deformed during the Late Cretaceous to Paleocene Laramide Orogeny (Price 1981).

Experimental Methods

Forty-seven oriented hand samples were collected - 11 from limestones, 23 from breccias and 13 from dikes - at individual sites (Fig. 2). The limestone samples represent about 20 m of stratigraphic thickness. Four specimens were cored from each sample, except for site 25 which yielded two, and their natural remanent magnetizations (NRM) were measured on an automated two-axis CTF cryogenic magnetometer. Four breccia sites (4,6,8 and 43) and one dike site (27) with weak NRM intensities ($<9 \times 10^{-5}$ Am^{-1}) were rejected. For the remaining 42 sites, one specimen was demagnetized with alternating fields (AF) in 11 steps to 100 mT using a Schonstedt GSD-1 AF demagnetizer and a second specimen was demagnetized thermally in 10 steps to 600°C using a Schonstedt TSD-1 thermal demagnetizer. Several diatreme specimens were further step demagnetized to 675°C. The remaining 82 third and fourth specimens were subjected to a three-step demagnetization, primarily by thermal cleaning, to isolate their stable remanence. The stable component was defined by the presence of a consistent direction over two or more steps accompanied by a significant decrease in intensity.

Remanence Components

Diatreme Breccia

AF step demagnetization gives poor results for the breccia specimens. In contrast, thermal step demagnetization isolates either one or two remanence components. Most specimens have a poorly-defined downwardly-directed component with unblocking temperatures (T_{ub}) of <400°C and a well-defined SE upwardly-directed component

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with T_{ub}'s from 500 to 675°C (Fig. 3a,b). The high temperature or reversed A component resides mostly in magnetite which unblocks between 500 and 600°C with a minor hematite contribution which unblocks above 600°C (Fig. 3a). In a few specimens (Fig. 3b), the low temperature or normal C component is well isolated and gives a NNW and steeply-downward remanence direction. In about one-third of the breccia specimens, only the A component is present (Fig. 3c). In these the direction is very stable and the T_{ub}'s indicate a predominance of hematite relative to magnetite.

The acquisition of isothermal remanent magnetization (IRM) shows that magnetite is the dominant magnetic mineral in the breccia specimens with most domains saturating at <0.2T (Fig. 4a). The increase of 1 to 3% between 0.3 and 0.9T is evidence that some hematite is also present, given that geothite can be ruled out from the AF and thermal demagnetization characteristics (Dunlop 1972, Lowrie and Heller 1981). Examination of the acquisition curves in more detail below 0.1T (Fig. 4b), indicates that the magnetite ranges from multidomain (MD) to pseudosingle domain (PSD) (i.e., small multidomain of ~0.2μm size). AF demagnetization of of the saturation IRM also indicates a MD to PSD size range (Fig. 4c). The curves for two additional breccia specimens fall between those shown in Fig. 4.

Three breccia sites were rejected after demagnetization. Two of these sites (31 and 37) had large α_{95} values while the remanence of the third site (20) proved too weak ($<9 \times 10^{-5}$ Am^{-1}) to be reliably measured. However a coherent A component remanence was found in 16 breccia sites (Table 1; Fig. 5a). When combined these sites give an overall mean direction of 135.3°, -37.1° (8.2°) (declination, inclination (α_{95})) (Fisher, 1953).

Diatreme Dikes

Except for two sites (11 and 32) at low 20 mT cleaning fields, AF demagnetization gave incoherent data for the dike sites. Thermal step demagnetization gave essentially the same results as found for the breccia specimens but with lesser stability overall. Most dike specimens have a very poorly-defined, downwardly-directed component with T_{ub}'s of <400°C and a well-defined SE upwardly-directed A component that resides in stable magnetite and hematite domains (Fig. 6a). One dike site (32) and a few other specimens have antiparallel A component directions with T_{ub}'s of >500°C (Fig. 6b). The response of the dike specimens to IRM analysis is essentially the same as found for the breccia specimens (Fig. 4).

AF and thermal demagnetization were successful in isolating a stable A component direction in all but two of the 12 demagnetized dike sites (Table 1; Fig. 5b). The two sites (1 and 12) had high α_{95} values associated with their mean remanence direction and were rejected. The mean remanence direction of the 10 dike sites is 137.5°, -37.4° (12.1°).

Diatreme Combined

The breccia and dike mean directions of 135.3°, -37.1° and 137.5°, -37.4° differ by only one degree or very much less than their

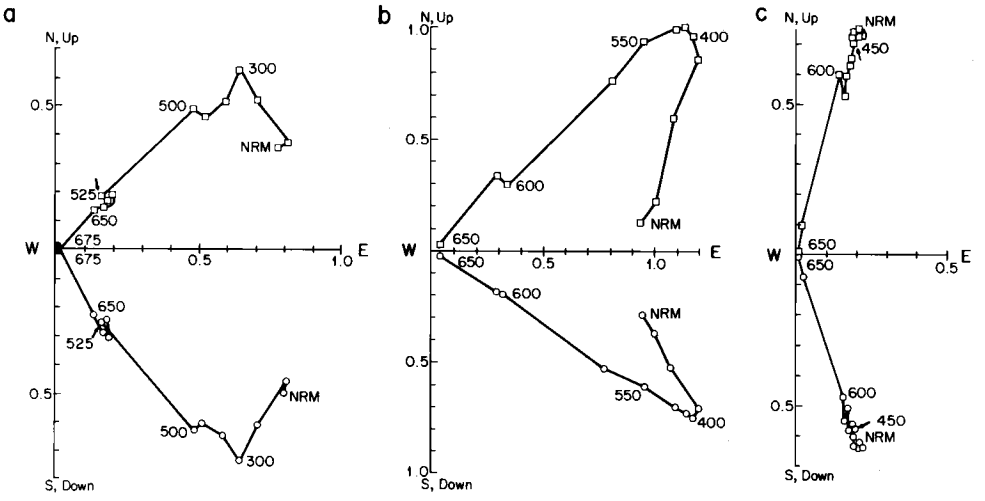


Fig. 3. Orthogonal vector demagnetization plots for diatreme breccia specimens from: a) site 30; b) site 19; and, c) site 21. Circles represent projections on the horizontal plane and squares on the vertical plane with solid (open) symbols denoting positive down (negative up) inclinations. The cleaning treatment values are in °C. The axial values are expressed as a ratio to the NRM intensity.

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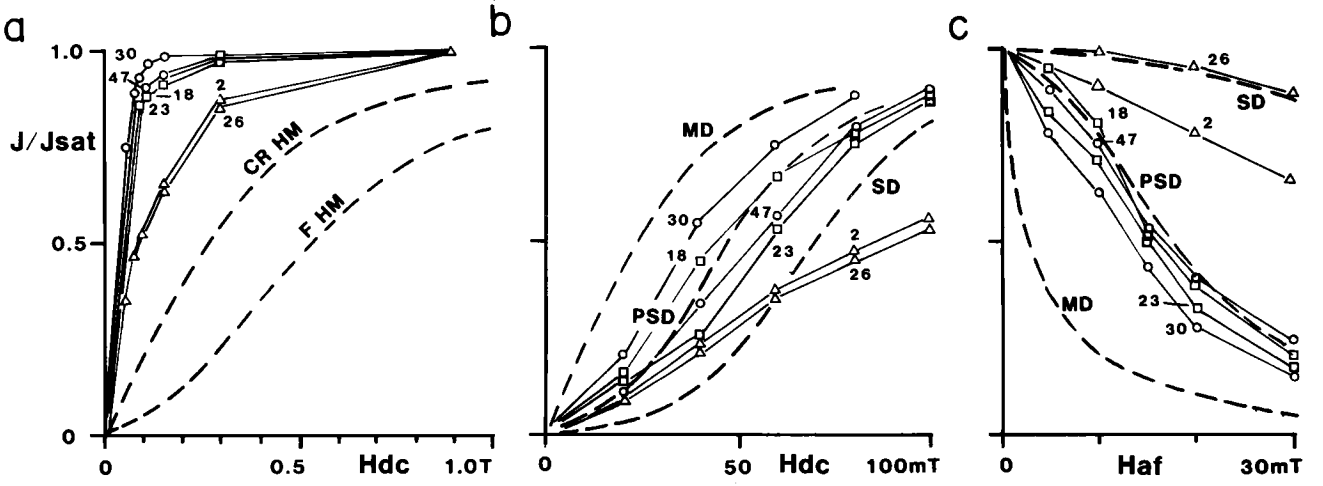


Fig. 4. Example IRM acquisition and demagnetization curves for breccia (circles), dike (squares) and limestone (triangles) specimens: a) acquisition of IRM with the dashed lines indicating coarse grained (CR) and fine grained (F) curves for hematite (HM) from Dunlop (1971); b) initial acquisition of IRM curves with multidomain (MD), pseudosingle domain (PSD, small multidomain of 0.2 μm size) and single domain (SD) curves for magnetite from Dunlop (1972); and c) AF demagnetization of saturation IRM (0.9T) with MD, PSD and SD curves from Dunlop (1973), 1981). The remanence intensities are expressed as a ratio of the saturation intensity (J/Jsat. The acquisition curves are plotted against the direct field intensity (Hdc) and the demagnetization curves against the alternating field intensity (Haf).

respective α_{95} values. Clearly they represent the same magnetization event with little difference in time, and can be validly combined into one population with a mean direction of 136.2°, -37.2° (6.5°). The concordancy of the massive dike direction with that of the foliated breccia phase also indicates that anisotropy of magnetic susceptibility associated with the foliation fabric in the breccia is not sufficiently strong to affect its remanence.

Host Rock Limestone

AF step demagnetization of the limestone specimens isolates a poorly-defined low-coercivity (<55 mT) component in nine sites that is directed NNW and steeply down (Fig. 7a). Between 450 and 550°C, thermal step demagnetization typically isolates a small residual B remanence component that is directed ESE with a shallow upward inclination (Fig. 7b). Noting the low T_{ub}'s it is probable that the limestone remanence is carried mostly by magnetite. The two tested limestone specimens responded very similarly to IRM analysis to give curves that lie virtually on top of each other in Fig. 4. The initial rapid increase in acquired IRM below 0.1T (Fig. 4a) indicates that magnetite is present and the slow increase from 0.3T indicates that some hematite may also be present. When looked at in more detail (Fig.

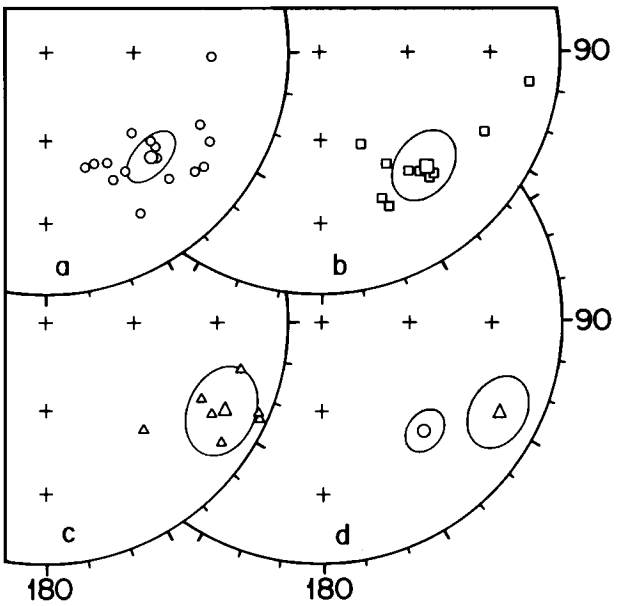


Fig. 5. Mean remanence directions shown on equal-area stereonet (upper hemisphere) segments for the a) diatreme breccia; b) diatreme dikes; c) host rock limestone; and d) diatreme combined (circle) and limestone (triangle). Small symbols represent site means and large symbols represent unit means with their cones of 95% confidence (Fisher, 1953).

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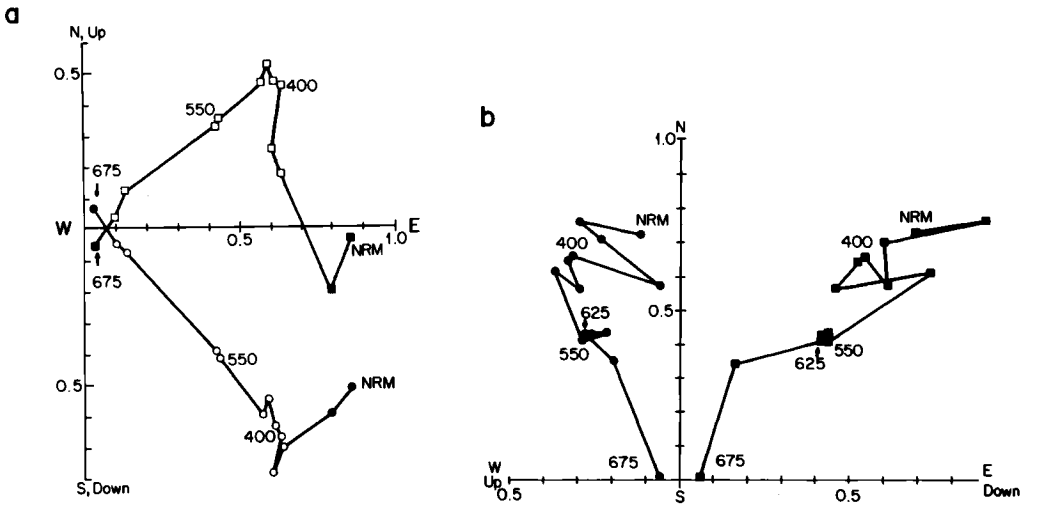


Fig. 6. Orthogonal vector demagnetization plots for diatreme dike specimens from: a) site 23; and b) site 32. Plotting conventions as in Fig. 3 except for "b" where the Up-N-Down-S vertical plane is used. Deviations of endpoint data from the origin are caused by background noise in the weakly magnetized specimens.

4b), it is evident that the magnetite is PSD or SD in character. Similarly AF demagnetization of the saturation IRM indicates PSD or SD behaviour (Fig. 4c). The remanence data from four limestone sites were rejected after demagnetization. Two of these sites (7 and 42) had unacceptably large α_{95} values associated with their mean directions, and two sites (16 and 41) retained dominant C component overprint. The reversed B component is carried by the seven remaining sites with a mean direction of 117.8°, -18.1° (12.4°) (Table 1; Fig. 5c).

As shown in Figure 5d, the cones of 95% confidence for the limestone B and diatreme A components do not intersect, and hence the two component directions are statistically significantly different at >>95% confidence. This result provides an imperfect contact test (Everitt and Clegg, 1962) that signifies that the A component of the diatreme is primary and the B component of the limestones predates intrusion of the diatreme. The test would be strengthened if a limestone sample from the contact had been collected and shown to be remagnetized to the A direction. For all seven accepted sites, the demagnetization directions track either northward or southward from the low coercivity NNW down direction to the reversed B direction; and then as cleaning approaches the magnetite Curie temperature, the directions swing very gradually to the northeast towards lower Cambrian polar positions. Thus it is clear that the B component is not simply a poorly-resolved A component or recent overprint.

The calculated cooling rate of the diatreme based on the equations of Carlslaw and Jaeger

TABLE 1. Site Mean Statistics

Site No.	N	Temp °C	R	Decl. °	Incl. °	α_{95}	k
A COMPONENT							
Breccia							
5	3	450-500	2.999	116.2	-30.4	2.6	2238
9	3	500-525	2.932	119.0	-23.9	23.1	30
10	2	500	1.975	157.8	-48.0	40.4	40
14	3	500	2.989	91.4	-32.4	9.2	182
15	3	500-575	2.987	152.9	-39.5	10.0	154
17	3	500	3.000	131.5	-42.2	1.0	13805
19	2	500	1.977	133.9	-36.5	38.7	44
21	3	500-575	2.997	161.9	-48.0	4.6	724
25	2	500	1.964	147.5	-40.0	49.1	28
28	3	500	2.997	152.0	-46.5	4.8	654
30	3	500	2.994	135.8	-27.8	6.5	360
36	3	500	2.964	126.6	-20.1	16.8	55
44	3	500-525	2.978	150.0	-23.7	13.0	91
45	3	500	2.954	131.2	-41.8	18.9	44
46	3	500	2.730	129.6	-22.1	49.0	7
47	3	500	2.990	143.3	-49.6	8.8	198
Mean (16)			15.300	135.3	-37.1	8.2	21
Dikes							
3	3	500-575	2.965	138.8	-32.5	16.3	58
11	3	20mT	2.850	115.0	-28.4	35.1	13
18	3	500	2.992	155.9	-32.9	8.0	236
23	3	500	2.992	137.6	-33.3	7.8	250
24	3	500-550	2.966	156.7	-36.0	16.1	60
32	3	20mT	2.998	143.5	-40.3	4.2	875
33	3	500-575	2.923	96.8	-13.6	24.7	26
35	3	500	2.987	156.3	-57.8	9.9	155
38	3	500	2.988	148.6	-46.6	9.6	167
40	3	500-575	2.784	140.4	-37.5	43.0	9
Mean (10)			9.469	137.5	-37.4	12.1	17
Breccia & Dikes							
(26)			24.766	136.2	-37.2	6.5	20
(26)		corrected	24.766	129.8	-44.3	6.5	20

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TABLE 1 (continued)

Site No.	N	Temp °C	R	Decl. °	Incl. °	α_{95}	k
B COMPONENT							
Limestone							
2	3	500	2.978	113.1	-4.0	13.0	91
13	3	525	2.875	114.4	-2.1	31.9	16
22	3	500	2.775	124.4	-12.3	44.0	9
26	3	500	2.987	116.4	-27.9	10.0	154
29	3	500	2.965	103.3	-16.8	16.4	57
34	3	500	2.995	119.0	-22.4	6.2	397
39	2	525	1.990	137.9	-39.5	24.3	108
Mean	(7)		6.754	117.8	-18.1	12.4	24
	(7)	corrected	6.754	114.4	-22.7	12.4	24
C COMPONENT							
Breccia							
	12	≤20mT					
	(7)	≤400	6.86	352.3	66.2	9.4	42
Dikes							
	15	≤20mT					
	(8)	≤400	7.87	352.3	70.4	7.7	53
Limestones							
	17	≤20mT					
	(8)	≤400	7.95	339.8	72.4	4.7	139
All above							
	44						
	(23)		22.64	348.4	70.0	3.9	61

Notes: N is the number of specimens (blocks) averaged; Temp. is the thermal cleaning temperature in degrees Celsius except for AF cleaning field in milliTesla (mT); R is the vector resultant; Decl. and Incl. are the declination and inclination of the mean remanence vector with the radius of its cone of 95% confidence in degrees (Fisher 1953); k is the precision parameter; and "corrected" indicates direction after simple tilt correction using the average attitude for the host rock limestones of N90°E 10°S (Pell 1986).

(1959) indicates that the magnetization was blocked over a period of about 10^2 years, thus probably not allowing for complete averaging of paleosecular variation (PSV). Conversely the expected value for PSV is about 13.2° ± 1.5° (Pullaiah and Irving, 1976) which compares closely with the 11.5° value found for this study and indicates that there was nearly sufficient time for the complete averaging of PSV. Both arguments suggest at the very best that the possible errors from insufficient averaging of PSV are small and cannot account for differences between either the diatreme or sediment poles or the diatreme and reference poles.

C Component

The normal C component is isolated quite clearly in the host rock limestones and in the breccia and dikes of the diatreme by vector subtraction using the NRM and low AF (≤20 mT) or thermal (≤400°C) step demagnetization data (Table 1). Because it is found in all three rock types in the low coercivity and unblocking temperature domains, it is evidently a post-intrusive overprint from a secondary metamorphic event and/or a recently acquired viscous remanent magnetization (VRM). It has an overall mean direction of 348.4°, 70.0° (3.9°).

Pole Positions

The poles for the C component in the three units are spread along the Late Cretaceous to present portion of the apparent polar wander path (APWP) of Irving and Irving (1982, using 20 Ma smoothing) (Fig. 8, Table 2). We interpret this to signify that the C component is a mixture of: a) remanence acquired during the dominantly normal-polarity Late Cretaceous epoch as the Laramide Orogeny impressed a minor metamorphic or uplift overprint on these rocks; and, b) VRM acquired during the past 0.7 Ma of the Brunhes Normal polarity chron.

The pole position for the primary A component of the HP Pipe is not so readily interpreted because of uncertainty in the Paleozoic APWP for North America. The Pennsylvanian-Permian (Kiaman Reversed Superchron) segment was well defined by Irving and Irving (1982) using a data base of more than twenty poles (Fig. 9b), and subsequent investigations have essentially confirmed or accepted this segment (e.g., Gordon et al. 1984, Irving and Strong 1985, Miller and Kent 1986a). Conversely the Devonian-Mississippian segment is less certain because several poles that were originally thought to be primary, have been shown to be secondary. Specifically the Mississippian Mauch Chunk, Terrenceville and Deer Lake red beds and the Devonian Catskill red beds and Onondaga limestone poles record a syn- to postfolding remanence of Kiaman age acquired during the Alleghanian Orogeny (Kent and Opdyke 1978 and 1985, Van der Voo et al. 1979, Kent 1982, Irving and Strong 1984, Miller and Kent 1986a,b). Most of the remaining ~50 poles from Devonian and Mississippian rocks fall close to the Kiaman APWP segment which has led to two hypotheses (Table 3, Fig. 9). The traditional hypothesis is that the Devonian-Mississippian segment about doubles back upon the Kiaman segment to form a hairpin akin to the Cretaceous 60 to 130 Ma hairpin (e.g., Kent and Opdyke 1978, Van der Voo 1981, Ellwood 1982, Seguin and Fyffe 1986). The alternative hypothesis is that virtually all the remaining poles come from rocks that were remagnetized during the Kiaman so that only poles with a demonstrated prefolding remanence should be used to define the Devonian-Mississippian segment (e.g., Roy and Morris 1983, Irving and Strong 1985, Miller and Kent 1986a). Supporters postulate several mechanisms of remagnetization. Irving and Strong (1985), noting that most of the poles come from units within ±10° of the Kiaman paleoequator,

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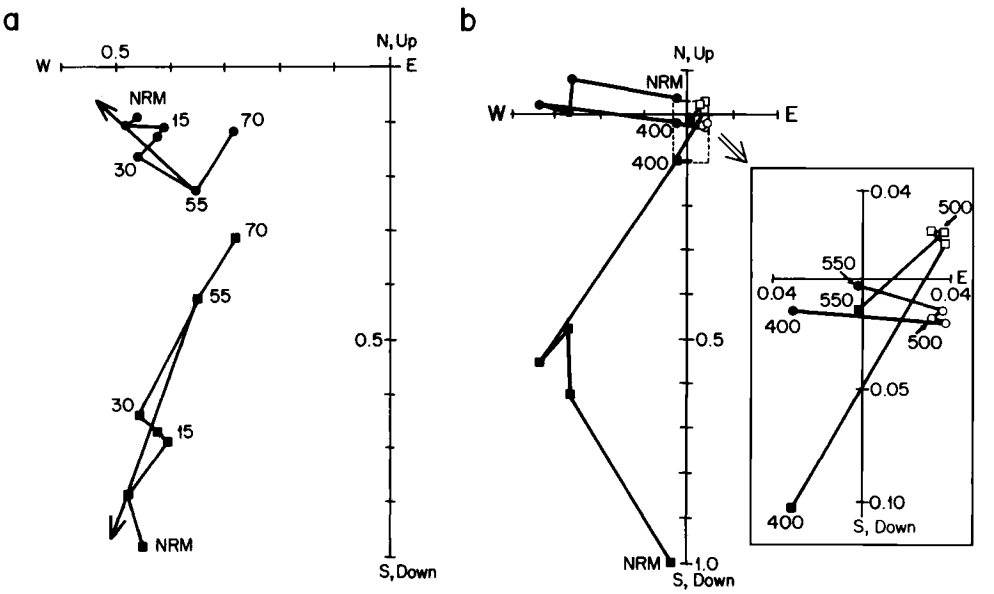


Fig. 7. Orthogonal vector demagnetization plots for host rock limestone specimens from: a) site 2; and b) site 26. Plotting conventions are as given for Fig. 3 except for "a)" where the treatment values are in mT and the arrows depict the low-coercivity C component.

suggest that deep tropical weathering could have formed hematite especially in the red beds. Following Hodych et al. (1984), they also suggest that primary goethite may have been converted to hematite by burial pressure in some units. Kent and Opdyke (1985), following McCabe et al. (1983), suggest that the growth of authigenic magnetite may have remagnetized some limestone units. Thermal overprints may also have been generated in rocks affected the Kiaman Alleghanian Orogeny.

If the remagnetization hypothesis is accepted then only three Mississippian (MC, DL, and TV; Table 3) and two Devonian (CA and CF) poles are demonstrably prefolding. The three Mississippian poles are on the south side of the main cluster of other Mississippian poles (Fig. 9a) but are not significantly different, and both Devonian poles are well within the Devonian population (Fig. 9b). The Devonian poles (CA and CF) come from the Catskill Formation and give significantly different positions because of relative rotation about the Pennsylvania Salient (Miller and Kent 1986a); which, then, is the correct Devonian position? The Mauch Chunk pole (MC) comes from the same part of Pennsylvania and is similarly suspect. The remaining two poles come from Newfoundland and may reflect the northward translation of Acadia (TV) or deformation associated with its impact on the craton (DL) (Miller and Kent 1986a). Certainly the Acadian poles overall are biased to the southeast side of both populations (Fig. 9a,b). Experimental

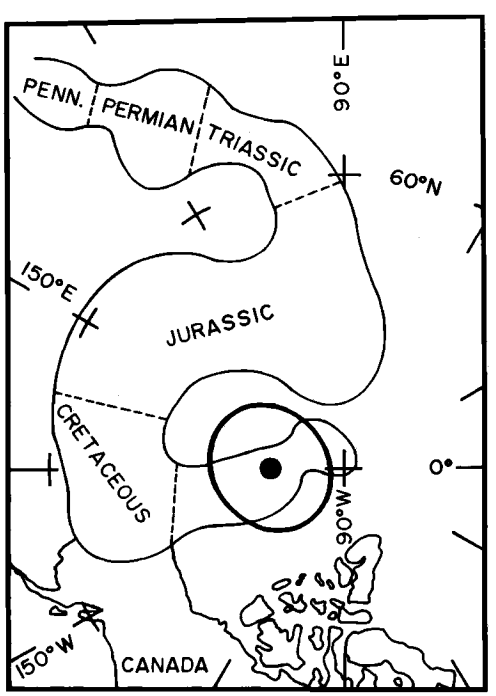


Fig. 8. North American APWP from Pennsylvania to present modified from Irving and Irving (1982, using 20 Ma window) with the pole for the C component circumscribed by its oval of 95% confidence.

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TABLE 2. Pole Positions for the HP Pipe Study

	Long. °E	Lat. °N	δ_p °	δ_m °
A Component - Diatreme				
26 sites	127.5	44.2	4.5	7.6
26 sites, tilt corrected	138.5	44.5	5.1	8.2
B Component - Limestones				
7 sites	136.4	24.3	6.7	13.0
7 sites tilt corrected	141.0	24.3	7.0	13.3
C Component				
Breccia, 7 sites	124.1	84.2	12.7	15.4
Dikes, 8 sites	188.7	84.5	11.6	13.4
Limestones, 8 sites	187.9	77.0	7.4	8.4
All above, 23 sites	175.0	82.7	5.8	6.7

Notes: The pole position is given by its easterly longitude, northerly latitude and semi-axes of its oval of 95% confidence (Fisher, 1953), all in degrees using the mean remanence directions as listed in Table 1.

uncertainties can also be raised. For example, pole CA averages data from only seven samples.

There are other concerns that the remagnetization hypothesis has not addressed well. The proposed mechanisms would require a high degree of timing coincidence to remagnetize all of the ~50 poles listed in Table 3, considering that they represent a full range of rock types from a wide spectrum of geologic environments throughout North America. The hypothesis must explain the progressive shift of the Mississippian, Upper and Lower Devonian directional clusters in spite of Kiaman resetting (Table 3). It also must explain why pole clusters such as found for the Traveler Terrane (poles BE, DO, CM, HQ and TF) (Brown and Kelly 1983) have not been reset. Nor is it reasonable to simply reject all plutonic poles because they lack tilt control. Most of these plutons are postorogenic so that tilt corrections are likely to be minor, and such tilt errors are likely to be randomized by the large number of plutons involved. The fact that only two such Mississippian (EG, SL) and three Devonian poles (MP, SS, WV) deviate significantly from their corresponding age clusters suggests this is true. Stratiform rock units may also have uncertain poles because of local tectonics. For example, the two Colorado Plateau poles (TB and

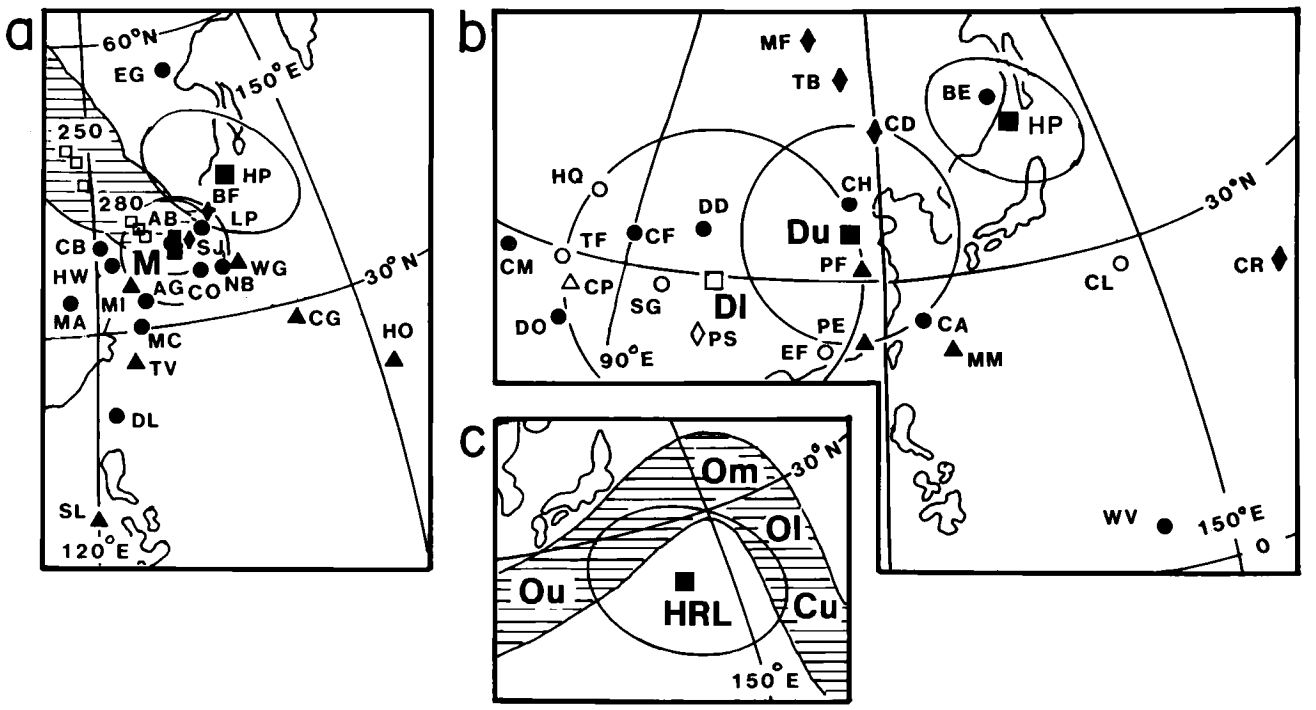


Fig. 9. North American pole positions as listed in Table 3 for: a) Mississippian time with the Kiaman track from Irving and Irving (1982); b) Devonian time separated into Early (open symbols) and Middle-Late (solid symbols) epochs; and c) the B component or host rock limestone pole of this study relative to the Cambro-Ordovician APWP of Van der Voo (1981). Poles from the Appalachians are shown as circles, from the Acadian Subterrane as triangles and from elsewhere in North America as diamonds with squares used to denote mean positions.

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TABLE 3. Devonian and Mississippian Pole Positions

Pole Code	Age	Unit and Lithology	Study Area	Tec	Pole Position			Reference
					Lat °N	Long °E	δ_p, δ_m or α_{95}	
HW	M	Hopewell Gp., red sediments	N.B.	O	37	122	2,3	Roy & Park 1969
MI	M	Minudie Point, red sediments	N.S.	A	35	124	3,6	Roy 1977
MC	M	Mauch Chunk, red beds	Penn.	O	31	125	8	Kent & Opdyke 1985
MA	M	Maringouin Fm, red sediments	N.B.	O	34	117	4,9	Roy & Robertson 1968
CO	M	Codroy Gp., red beds	Nfld.	O	36	133	5,10	Black 1964
LP	M	Lilesville pluton, gabbro/granite	N.C.	O	40	134	3	Barton & Brown 1983
WG	M	Windsor GP., red sediments	N.S.	A	36	137	3,6	Scotese et al. 1984
CG	M	Canso Gp., red sandstone	N.S.	A	29	143	12,22	Scotese et al. 1984
BF	M	Barnett Fm, limestone/concretions	Texas	C	41	135	4,8	Kent & Opdyke 1979
HO	M	Horton Gp., red sandstone	N.S.	A	24	152	3,6	Spariosu et al. 1984
DL	M	Deer Lake Gp., red/grey clastics	Nfld.	O	22	122	6,10	Irving & Strong 1984
NB	M	New Brunswick basalts	N.B.	O	36	136	3,6	Seguin & Fyffe 1986
SJ	M	St. Joe Limestone	Ark.	C	39	132	3	Scott 1979
SL	M	St. Lawrence granite alaskite/dikes	Nfld.	A	12	120	9,13	Irving & Strong 1985
AG	M	Austell granitic orthogneiss	Georg.	O	34	126	7,13	Ellwood & Abrams 1982
HP	M	HP Pipe, kimberlite/diabase	B.C.		45	139	5,8	this study
TV	M	Terrenceville Fm., red sandstone	Nfld.	A	27	124	6,11	Kent 1982
CB	M	Charlotte Belt gabbros, 5 plutons	S.C.	O	39	121	10	Dooley 1983
AR	M	Atlanta granites, 6 plutons	Geor.	O	39	130	11,18	Ellwood 1982
EG	M	Elberton granites, 6 plutons	Georg.	O	56	134	10,14	Ellwood 1982
CM	Du	Compton metasediments	Que.	O	28	77	4,8	Seguin et al. 1982
DO	Du	Dockendorff Gp., basic volcanics	Maine	O	24	84	7	Brown & Kelly 1980
TB	Du	Temple Butte Fm., limestone	Ariz.	C	53	115	1,2	Elston & Bressler 1977
MF	Du	Martin Fm., red beds	Ariz.	C	56	109	1,2	Elston & Bressler 1977
CD	Du	Columbus & Delaware limestone	Ohio	C	46	120	2,3	Martin 1975
CR	Du	Columbus & Raisin R. dolomite	Ohio	C	24	164	9,11	Martin 1975
CF	Du	Catskill Fm, red beds	Penn.	O	33	90	7	Miller & Kent 1986a
CA	Du	Catskill Fm, red beds	Penn.	O	26	124	15	Miller & Kent 1986b
BE	Du	Belchertown pluton, monzonite	Mass.	O	48	147	6,13	Ashwal & Hargrave 1977
MP	Du	Mt. Peyton batholith, granite	Nfld.	O	-63	125	14	Lapointe 1979
PF	Du	Perry, Fm, red beds	N.B.	A	32	118	8,16	Robertson et al 1968
PE	Du	Perry, Fm., volcanics	NB,ME	A	25	118	10,17	Black 1964*
CH	Du	Charlotte Belt "old" granite	NC,SC	O	39	116	-	Ellwood 1982
WV	Du	Wesleyville dikes, diabase	Nfld.	O	4	144	9,13	Murthy 1983b
DD	Du	Deadman's Bay dikes, diabase	Nfld.	O	35	98	5,10	Murthy 1983a
MM	Du	Massachusetts mafic complex	Mass	A	23	126	10	Schutts et al. 1976
SG	Dl	St. George pluton, gabbro	N.B.	O	29	93	9,15	Roy et al. 1979
SS	Dl	St. Stephen Pluton, gabbro	N.B.	O	-43	135	35,35	Roy et al. 1979
CL	Dl	Clam Bank Gp., red beds	Nfld.	O	28	146	6,12	Black 1964
PS	Dl	Peel Sound Fm., red beds	Arctic		25	99	5,10	Dankers 1982
DA	Dl	Dalhousie Gp., volcanics/diabase	Que.	O	- 1	68	7,9	Seguin & Gahé 1984
TF	Dl	Traveler felsite	Maine	O	29	82	4,8	Spariosu & Kent 1983
EF	Dl	Eastport Fm, red beds/lavas	Maine	O	24	114	7,11	Kent & Opdyke 1980
CP	Dl	Cape Breton granites	N.S.	A	27	84	6,11	Rao et al. 1981
HQ	Dl	Hartwick quarts diorite	Mass.	O	36	84	-	Brown & Kelly 1977
	M	Mean - all 20 poles			35	130	5	k = 48
		- 17 poles excluding EG,HO,SL			35	131	4	k = 70
	DU	Mean - 15 poles excluding MP			35	117	12	k = 12
	Dk	Mean - 7 poles excluding DA,SS			30	100	16	k = 16

Notes: For Age M is Mississippian, Du is Middle and Late Devonian and Dl is Early Devonian. The tectonic regime is A for Acadia, O for orogenic and C for cratonic.

*See also Phillips and Heroy 1966.

MF) are on the north side of the Devonian cluster of poles, consistent with the ~9° of clockwise rotation proposed by Steiner (1986). Similarly red beds are perhaps the most likely rock type to have been pervasively reset. Yet when these poles are examined after excluding those shown to be prefolding, they are biased to the south side of the Mississippian cluster (CG, HOO HW, MA, MI and WG in Fig. 9a) or randomly distributed in the Devonian cluster (MF, PS, EF, PF and CL) rather than biased to the north as remagnetization implies. Thus we hold that no pole should be automatically categorized as "remagnetized" simply because of its direction but rather it must first be shown to be secondary by field tests at least until a

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coherent cluster of pre-folding poles is defined from different parts of the stable craton for each era. On this premise the Mississippian and Devonian poles have been calculated using all of the data (Table 3).

The pole position for the primary A component of the HP Pipe after tilt correction overlaps directly on the Mississippian portion of the APWP (Fig. 9a). This is consistent with the Mississippian Rb-Sr age of 348 ± 7 Ma and the previously stated interpretation that the Early Devonian K-Ar ages of 391 ± 12 and 396 ± 10 Ma contain excess initial argon. When the HP Pipe pole is not corrected for the 10°S dip, then the pole shifts slightly into the Pennsylvanian portion of the APWP. This result supports the geologic evidence that the tilting occurred after acquisition of the A component, probably during Laramide time. The mean Mississippian pole position of 131°E, 35°N is about ten degrees away from the HP Pipe pole, but is not statistically significantly different. This result means that the HP Pipe has probably not been significantly translated or rotated relative to the craton since emplacement in Mississippian time. Even if one accepts the remagnetization hypothesis, there is little change in the tectonic conclusions of this study i.e., the study area would be cratonic since Kiaman instead of Mississippian time.

The pole position for the B component of the host rock limestones straddles the Late Cambrian to Middle Ordovician portion of the APWP of Van der Voo (1981) after correction for the 10°S tilt (Fig. 9c). This result agrees directly with the assigned stratigraphic age. It indicates that the B component isolated in the limestones which is constrained to being pre-Mississippian by the contact test, is a primary or early diagenetic remanence. Finally, this result also means that the host rock limestones have probably not been rotated or translated relative to the craton since deposition in about Early Ordovician time.

Discussion and Conclusion

The Rocky Mountain fold-and-thrust belt trends NW along a great-circle that passes through eastern China and relatively close to the measured pole positions. This is a convenient geometry because a significant translation of an allochthonous Cordilleran terrane along the western margin of the North American craton should be directly observable as a change in paleolatitude. Noting that HP Pipe pole is about ten degrees near-sided, equivalent to ~1000 km of southward translation of an allochthon, and that the host rock limestone pole is about the same amount far-sided or equivalent to ~1000 km of northward translation, the simple conclusion from this data is that there has not been significant NW-SE translation of the HP Pipe area relative to the craton. If translation were postulated for the pipe then it is opposite in direction to the northward motions observed in outboard units. The data does not constrain SW-NE compressional translation during the Laramide or some earlier orogeny.

The geometry is also convenient because the measured and reference pole positions range from 69° to 79° distant from the sampling site. This means that a small rotation of the HP Pipe around a vertical axis would be obvious. For example, correcting the pole for a 5° counterclockwise rotation of the pipe would slightly improve the good fit of the host rock limestone's pole. Conversely this rotation would distinctly degrade the good fit of the HP Pipe pole. Thus these results constrain the relative rotation between the pipe and craton to being ≦5°. Such a minor amount argues strongly against the idea that the pipe is part of an accreted allochthon which would normally show a significant rotation. It also indicates that the ~200 km of shortening associated with the SW-NE Laramide compression in the Golden map-area (Price 1967, 1981) did not result in a rotation of the thrust slice housing the HP Pipe as it moved NE onto the craton.

The finding of concordant pole positions for both the Mississippian HP Pipe and its Late Cambrian to Middle Ordovician host rock limestones confirms that this part of the Rocky Mountain fold-and-thrust belt has been part of the North American craton during Phanerozoic time. Thus the suture at the craton's western margin must be outboard of the study area in the Rocky Mountain Trench as Stone (1977) and Chamberlain and Lambert (1985) suggest, at the eastern edge of Terrane I as Monger et al. (1982) suggest, or distributed between the Trench and Terrane I on several faults as Irving et al. (1985) suggest.

The HP Pipe result is most notably at variance with the results from the Early Cretaceous Crowsnest volcanics that Irving et al. (1986) reported. This unit crops out on the inboard side of the fold-and-thrust belt to the southeast of the study area (CV in Fig. 1), and its remanence suggests 17 ± 6° of northward translation and 24 ± 10° of clockwise rotation. Uncomfortable with this geotectonic explanation for geologic reasons, Irving et al. (1986) prefer the explanation that the unit was extruded extremely rapidly so that secular variation of the EMF was not averaged out in the result. Our results support their preferred explanation. They are also consistent with the results of Grubbs and Van der Voo (1976) from Triassic red beds in the fold-and-thrust belt of Idaho-Wyoming. They found no evidence of translation in the measured inclinations and relate discordant declinations to local deformation of overthrust sheets around foreland salients. Paleomagnetic data from younger Jurassic and Cretaceous formations in the thrust sheets of this same area are also reasonably interpreted

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without invoking an allochthon model for the fold-and-thrust belt (McCabe et al. 1982; Schwartz and Van der Voo 1984). Recently Park and Aitken (1986a,b) have summarized the results from several paleomagnetic studies in the fold-and-thrust belt of the McKenzie Mountains in the Northwest Territories (MM in Fig. 1). These Late Proterozoic rocks have yielded a variety of apparently discordant poles that tempted geotectonic explanations. Park and Aitken (1986a,b) now believe that the lower latitude poles define a new loop in the APWP for the 0.9 to 0.7 Ga time interval, and that the high latitude poles record post-Paleocene secondary alteration. Thus there is no unequivocal paleomagnetic evidence for placing the western margin of the North American craton any further east than the Rocky Mountain Trench.

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